

Lab 3: Architecture Selection & Justification

System: Pet Adoption And Shelter Management System

Architecture Selection: Layered Architecture was chosen for the Pet Adoption Network.

Reason 1: Clear Separation of Concerns Across a Cohesive Feature Set

The system's functionality browsing available pets, submitting adoption applications, authenticating users, and verifying medical/vaccination records forms one cohesive application rather than independently scaling services. A layered structure (Presentation, Business, Data) lets the User Interface, Pet Catalog, Adoption Application Manager, Authentication, and Medical Records components evolve independently within the Business layer while sharing a single, consistent Database layer, without the operational overhead of managing distributed services.

Reason 2: Simplifies Data Consistency for Applications and Medical Checks

Core use cases (submitting an adoption application, confirming a pet's vaccination status before approval) depend on strongly consistent reads of pet availability and medical records for example, preventing two applicants from being approved for the same pet, or approving an application before its medical check has cleared. A single, centralized Data layer avoids the eventual-consistency and cross-service coordination challenges that a microservices split would introduce for this workflow.

Security Advantage

The Authentication component sits entirely within the Business layer and is the sole gatekeeper for verifying a user's identity before any adoption action is taken. Because the Presentation layer has no direct access to the Data layer, the User Interface cannot bypass login verification to read or modify application or medical data directly. Every request is forced through Authentication's identity check, reducing the attack surface for unauthorized account access or application tampering.

Performance Benefit

Isolating Medical Records within the Business layer, with a single required interface to the Database, allows vaccination-status lookups to be optimized and, if needed, cached independently of the Presentation layer. This lets the Adoption Application Manager confirm a pet's medical clearance quickly during the approval flow, without requiring changes to the User Interface or the Data schema as the volume of listings and applications grows.